

UNIT - 2

Introduction to C

HISTORY OF C

C was developed in 1972 by Dennis Ritchie at Bell Laboratories .

C was initially developed for writing system software.

Today, C has become a popular language and various software programs are written using this language.

Characteristics of C/Features

A high level programming language

Small size. C has only 32 keywords. This makes it relatively easy to learn

Makes extensive use of function calls

Stable language

Quick language

Characteristics of C/Features

C is a core language

C is a portable language

C is an extensible language

USES OF C

C language is primarily used for system programming. The portability, efficiency, the ability to access specific hardware addresses and low runtime demand on system resources makes it a good choice for implementing operating systems and embedded system applications.

STRUCTURE OF A C PROGRAM

- 1. Documentation Block**
- 2. Preprocessor Directives Block**
- 3. Global Declarations Block**

STRUCTURE OF A C PROGRAM

4. Main Block

```
main()  
{  
    Local declarations  
    Statements  
}
```

5. Function Block

```
Function 1()  
{  
    Local  
    declarations  
    Statements  
}
```

```
Function N()  
{  
    Local  
    declarations  
    Statements  
}
```

STRUCTURE OF A C PROGRAM

1. Documentation Block

This block is also known as comment block in which programmer can write definition of program and other details related to that program. This part is an optional.

There are two ways to insert comment in C program.

(1) Single line Comment using //

(2) Multi line Comment using /* */

STRUCTURE OF A C PROGRAM

2. Preprocessor Directives Block

This block contain special instructions that indicate how to prepare program for compilation.

One of the most important and commonly used preprocessor command is *include* which tells the compiler that to execute the program, some information is needed from the specific header file.

STRUCTURE OF A C PROGRAM

3. Global Declarations Block

This block is used to declare all global variables which will be accessible in an entire program / in all the functions. if you declare any variable out side the main function, it is considered as a global variable.

STRUCTURE OF A C PROGRAM

4. Main Block

The main() function is most important function and is a part of every C program. The execution of a C program begins at this function.

STRUCTURE OF A C PROGRAM

5. Function Block

A function is a subprogram or block of statement which performs a particular task.

Generally there are two types of function.

- 1. Library or Built in functions**
- 2. User Defined Function**

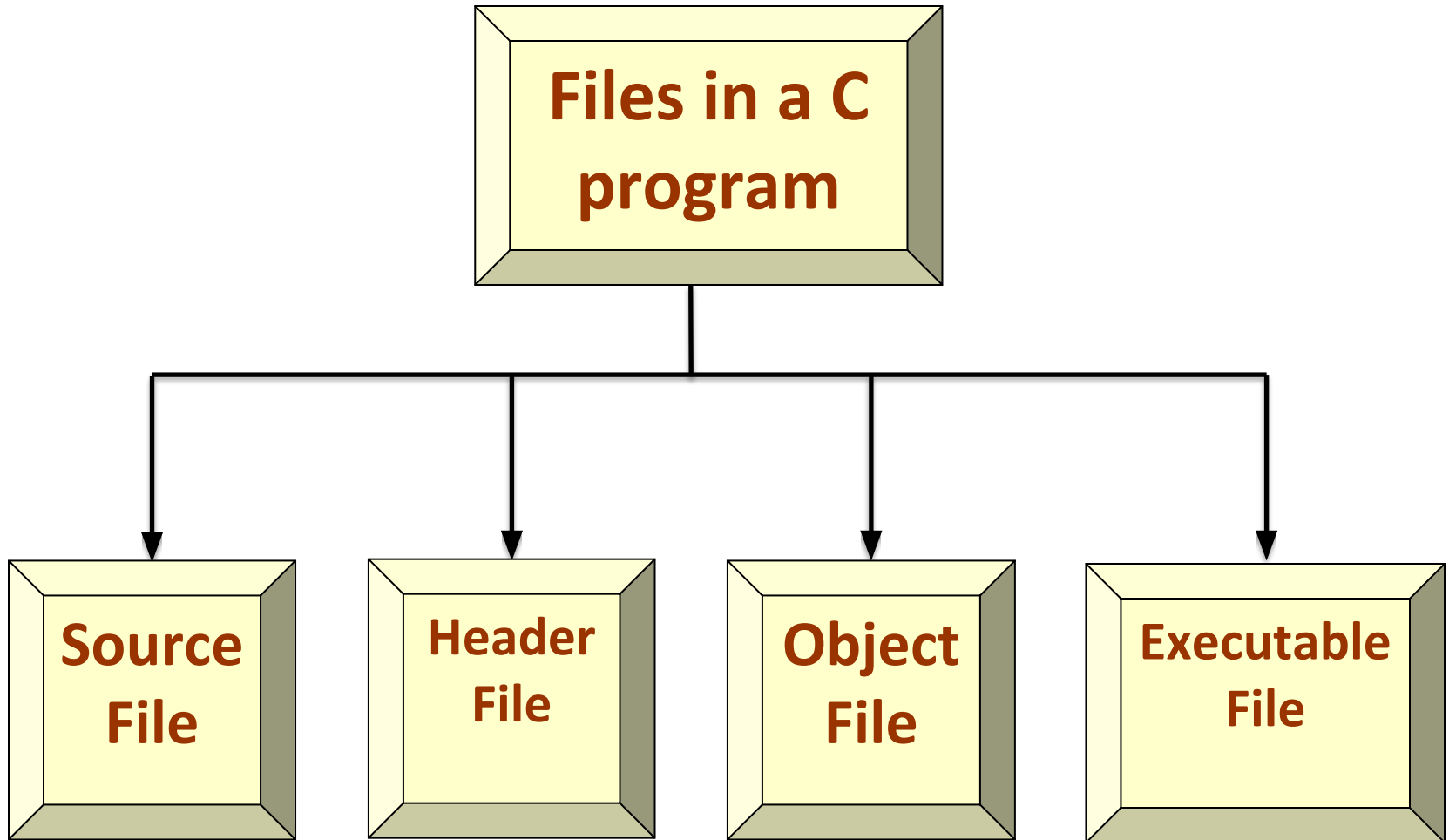
All user defined functions are declared out side the main function.

YOUR FIRST C PROGRAM

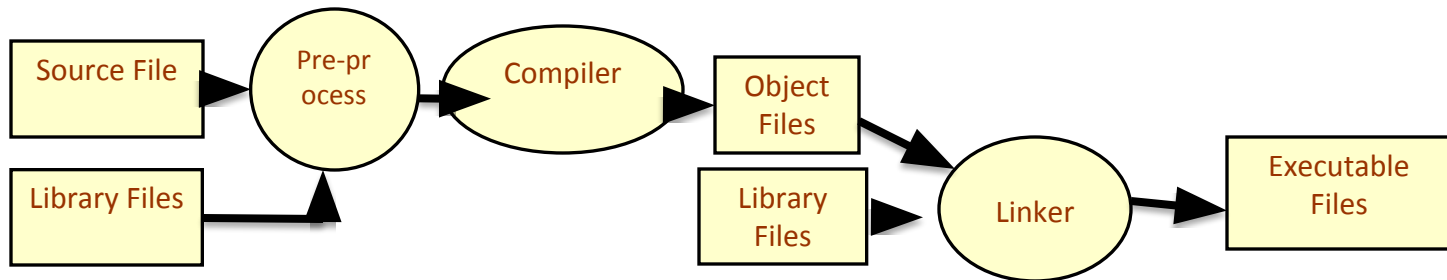
// Program that display Welcome Message

```
#include<stdio.h>
#include<conio.h>
void main()
{
    clrscr();
    printf("\n Welcome to the world of C ");
    getch();
}
```

FILES USED IN A C PROGRAM



COMPILING AND EXECUTING C PROGRAMS



Tokens

In C programs, each word and punctuation is referred to as a token. C Tokens are the smallest building block or smallest unit of a C program.

Following are considered as tokens.

1. Keywords
2. Identifiers
3. Constants

KEYWORDS

C has a set of 32 reserved words often known as keywords. All keywords are basically a sequence of characters that have a fixed meaning. By convention all keywords must be written in lowercase (small) letters.

KEYWORDS

32 Keywords are as follows.

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while

IDENTIFIERS

Identifiers are names given to program elements such as variables, arrays and functions.

RULES FOR NAMING IDENTIFIER

- it cannot include any special characters or punctuation marks (like #, \$, ^, ?, ., etc) except the underscore "_".
- Keywords cannot be used as identifiers

RULES FOR NAMING IDENTIFIER

- The names are case sensitive. So, example, “FIRST” is different from “first” and “First”.
- It must begin with an alphabet or an underscore.
- It can be of any reasonable length. Though it should not contain more than 31 characters.
- Example: Roll_no,marks,HRA,DA,etc.

DATA TYPES

Data types are used to store various types of data that is processed by program.

Data type attaches with variable to determine the number of bytes to be allocated to variable and valid operations which can be performed on that variable.

C Provides a standard, minimal set of basic data types. Some times these are called primitive data types.

BASIC DATA TYPES

Data Types	Keyword Used	Size in Bytes	Format Specifier	Range	Use
Character	char	1	%c	-128 to 127	To Store Characters
Integer	int	2	%d	-32768 to 32767	To Store integer Numbers
Floating Point	float	4	%f	3.4E-38 to 3.4E+38	To Store floating point numbers
Double	double	8	%f	1.7E-308 to 1.7E+308	To store big floating point numbers
Valueless	void	0	-----	Valueless	-----

DATA TYPES IN C

In addition, C has four type of qualifiers, also known as type modifiers, which preceded the basic type. They are as follows.

short

long (to increase the size of int or double type)

signed

unsigned

DATA TYPES IN C

However, the void does not have these modifiers. These type modifiers changes the meaning of the basic type.

The qualifier is classified into two types

(1) Size qualifier , e.g. short and long

(2) Sign qualifier , e.g. signed and unsigned

DATA TYPES IN C

DATA TYPE	SIZE IN BYTES	RANGE
char	1	-128 to 127
unsigned char	1	0 to 255
signed char	1	-128 to 127
int	2	-32768 to 32767
unsigned int	2	0 to 65535
signed short int	2	-32768 to 32767
signed int	2	-32768 to 32767
short int	2	-32768 to 32767
unsigned short int	2	0 to 65535
long int	4	-2147483648 to 2147483647
unsigned long int	4	0 to 4294967295
signed long int	4	-2147483648 to 2147483647
float	4	3.4E-38 to 3.4E+38
double	8	1.7E-308 to 1.7E+308
long double	10	3.4E-4932 to 1.1E+4932

VARIABLES IN C

A variable is defined as a meaningful name given to the data storage location in computer memory.

When using a variable, we actually refer to address of the memory where the data is stored. C language supports two basic kinds of variables.

- 1. Numeric variables**
- 2. Character Variables**

VARIABLES IN C

1. Numeric variables

It can be used to store either integer values or floating point values.

While an integer value is a whole numbers without a fraction part or decimal point, a floating point number, can have a decimal point in them.

VARIABLES IN C

2. Character variables

It can include any letter from the alphabet or from the ASCII chart and numbers 0 – 9 that are put between single quotes.

VARIABLES IN C

Declaring Variables :

Each variable to be used in the program must be declared.

To declare a variable, specify the data type of the variable followed by its name.

Example: int a;

VARIABLES IN C

Initializing Variables :

While declaring the variables, we can also initialize them with some value.

Example :

```
int roll_no=5;
```

RULES OF VARIABLE DECLARATION

All Rules for Naming Identifier are also rules for Variable Declaration.

To declare a variable specify data type of the variable followed by its name.

Variable names should always be meaningful and must reflect the purpose of their usage in the program.

Variable declaration always ends with a semicolon.

CONSTANTS

Constants are identifiers whose value does not change.

Constants are used to define fixed values like PI so that their value does not get changed in the program even by mistake.

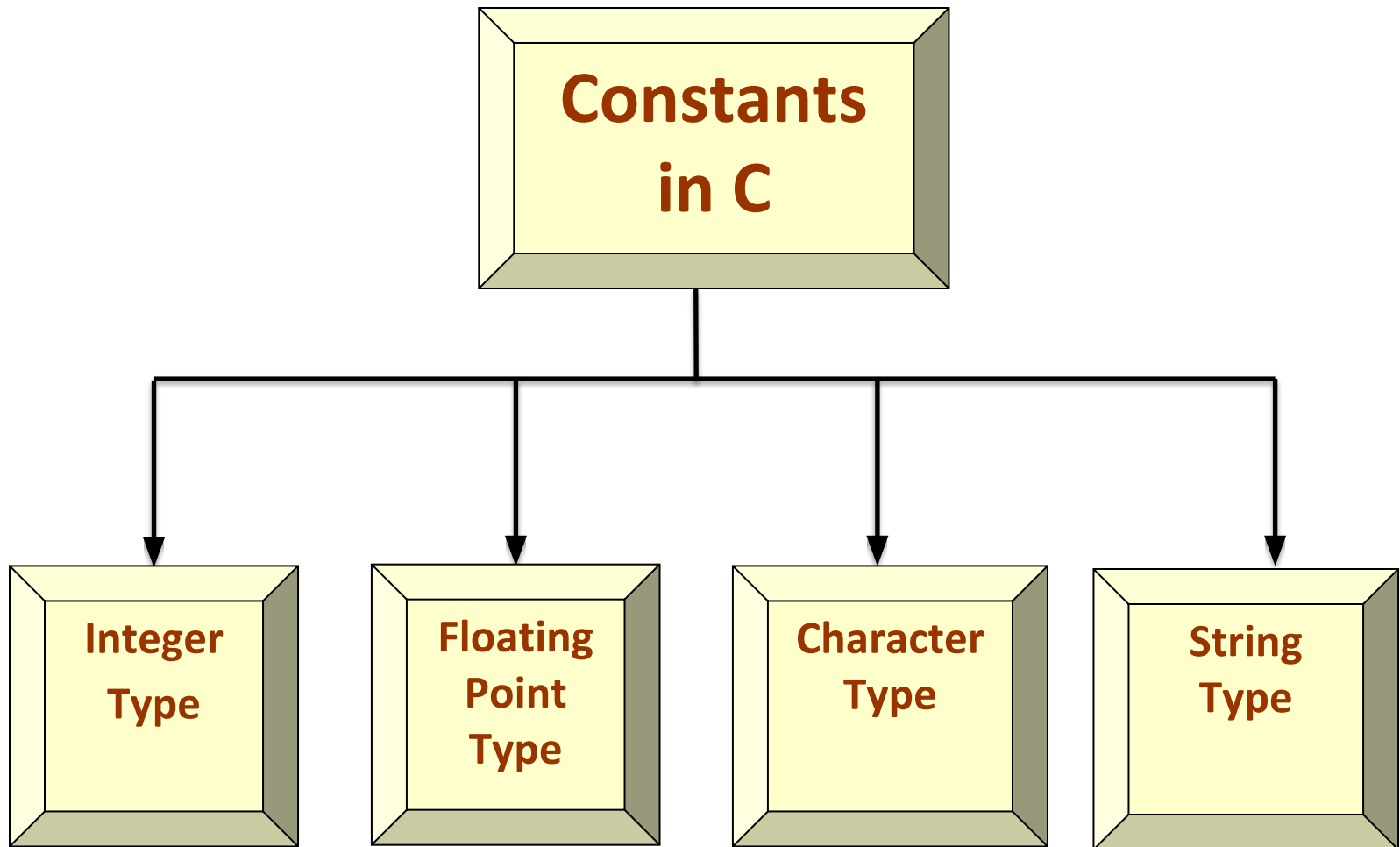
CONSTANTS

To declare a constant, precede the normal variable declaration with `const` keyword and assign it a value.

For example,

```
const float pi = 3.14;
```

CONSTANTS



CONSTANTS

1. Integer Constant

A constant of integer type consist of a sequence of digits.

For Ex:

1234 is of type int by default.

1234567L is valid long integer constant('L' or 'l')

12U is valid unsigned int constant('U' or 'u')

CONSTANTS

2. Floating Point Constant

Integer numbers are inadequate to express numbers that have a fraction part.

A Floating point constant therefore consist of an integer part, a decimal point , fraction part and an exponent field containing e or E(e means exponent)

For Ex: 0.02, +0.34,0.02E,etc.

CONSTANTS

3. Character Constant

A character constant consists of a single character enclosed in single quotes.

For example : 'a' , '\$' are character constants.

CONSTANTS

4. String Constant

A String Constant is a sequence of characters enclosed in double quotes.

The characters comprising the string constant are stored in successive memory locations.

For Ex: “Hello”

CONSTANTS

Declaring Constants in other way

(Defining Symbolic Constant)

Another way to designate a constant is to use the pre-processor command *define*.

```
#define PI 3.14159
```

When the preprocessor reformats the program to be compiled by the compiler, it replaces each defined name with its corresponding value wherever it is found in the source program. Hence, it just works like the Find and Replace command available in a text editor.

CONSTANTS

Rules that needs to be applied to a #define statement which defines a constant.

Constant names are usually written in capital letters to visually distinguish them from other variable names which are normally written in lower case characters. Note that this is just a convention and not a rule.

CONSTANTS

No blank spaces are permitted in between the # symbol and define keyword

Blank space must be used between #define and constant name and between constant name and constant value

#define is a pre-processor compiler directive and not a statement. Therefore, it does not end with a semi-colon.

Escape Sequence Characters

\a	Alarm or Beep
\b	Backspace
\f	Form Feed(Page Breaking in Print)
\n	New Line
\r	Carriage Return
\t	Tab (Horizontal)
\v	Vertical Tab
\\	Backslash
\'	Single Quote
\"	Double Quote
\0	Null

THANK YOU